

App Dev Project Report

1. Student Details

Name: Aman Raj

Roll Number: 24F3002906

Email: 24f3002906@ds.study.iitm.ac.in

2. Project Details

Project Title: Trekking Management (TrekOS)

Problem Statement:

To design and build a web-based application that allow users to Explore Treks and Book in the website.

Approach:

The app was built using Flask as the backend framework with a modular structure. It allows users to Book their trek trips.

3. AI/LLM Declaration

I used Google Gemini for debugging and creating API documentation samples, and improving variable naming consistency. I used 15-18%.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
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Flask	Core backend web framework
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SQLAlchemy	Object Relational Mapper for SQLite database
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Jinja2	Template engine for rendering dynamic HTML pages
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CSS	Frontend styling and responsive design
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SQLite	Lightweight local database for storing user data
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5. Database Schema / ER Diagram

Tables:

1.Users : stores user profile details (id, name, email,...etc.)

2.Trek : stores trek details (Location, trek name,...etc.)

3.Booking: stores detail of booking (user_id, trek-id, status,..etc.)

4.Staff: stores staff details (id, user_id, experience,...etc.)

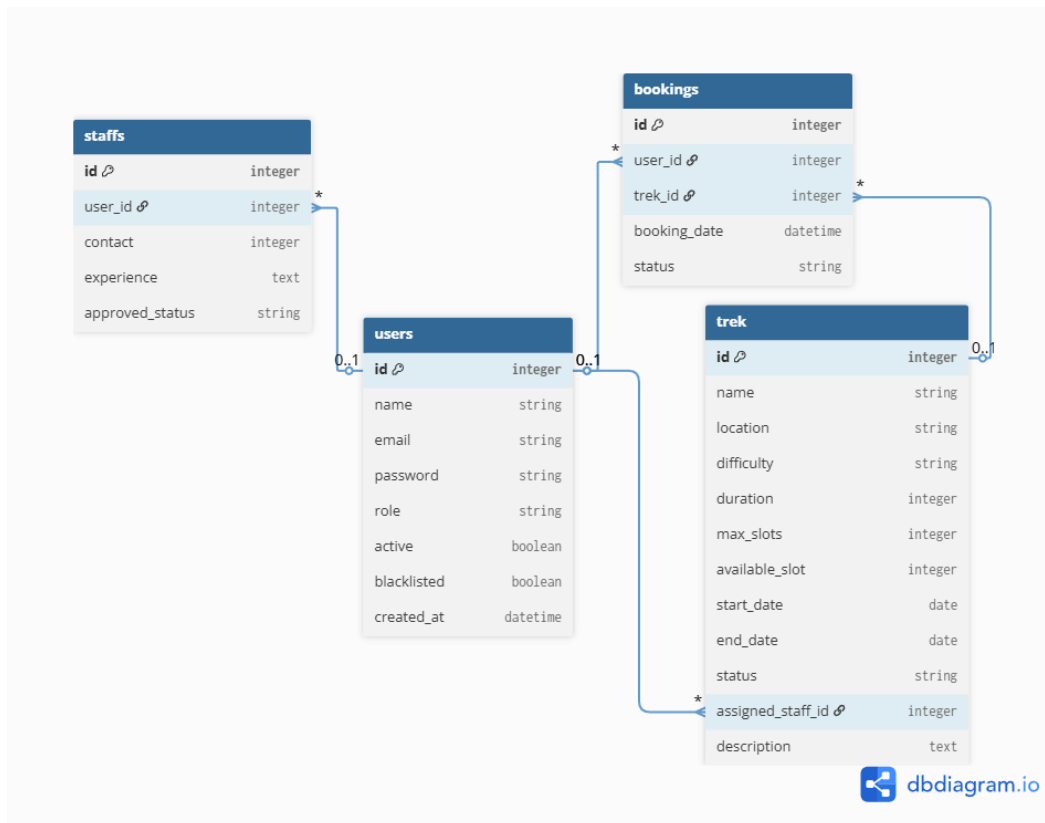
Relationships:

User (staff) ——— Trek [One staff can have many treks]

User (trekker) ———Booking [One user can have many bookings]

Trek —————Booking [One trek can have many bookings]

User (staff) ———Staff [One-to-one]



6. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/templates** – Jinja2 HTML templates
- **/static** – CSS
- **Config.py** -- Configuration (secret key, DB path)

7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1G-I2xFAT-JcqrD4VBjpdVeidTI7ujYxW/view?usp=sharing>